Fundamental Units

Exam code: 9MA0

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Fundamental units

What are fundamental units?

- **Fundamental units** or **S.I. units** are the international standardised units used around the world
- The three main units you will be using are:
  - **Length** measured in **metres (m)** (also called displacement)
    - 1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm
  - **Time** measured in **seconds (s)**
    - 1 hour = 60 minutes, 1 minute = 60 seconds
  - **Mass** measured in **kilograms (kg)**
    - 1 kg = 1000 g, 1 g = 1000 mg
- All other quantities can be measured in units **derived** from S.I. units

Help me understand

Worked Example

Convert the following into S.I. units:

a) 27.8 km

b) 3.8 hours

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

a) SI UNIT FOR LENGTH IS METRES (m)

27.8 km × 1000 = 27 800 m

1 km = 1000 m

b) SI UNIT FOR TIME IS SECONDS (s)

3.8 hours × 3600 = 13 680 s

1 HOUR = 60 MINS x 60 SECONDS = 3 600 SECONDS

c) SI UNIT FOR MASS IS KILOGRAMS (kg)

0.054 GRAMS ÷ 1000 = 0.000054 kg = 5.4×10^{-5} kg

1 kg = 1000 GRAMS

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Examiner Tips and Tricks

- Make sure you fully understand the definitions of all the words in this section so that you can be clear about what your exam question is asking of you.
- Make sure you are confident converting between fundamental S.I Units
  e.g. 17 minutes and 42 seconds = 1062 seconds

  2.4 km = 2400 m

  86000 mg = 0.086 kg.
- Watch out for exam questions using non S.I. units such as feet and inches or miles per hour.
- Remember mass (measured in kg) is not a weight (a force measured in Newtons).

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